Team No. 198 M2 – Progress

Green Customer Insights Platform

PS1 · Lloyds Banking Group · Megathon X: Singularity

1 Where we are

Every stage now runs end to end on a calibrated synthetic bank of **30,000 customers and 28.3M transactions**, on one laptop (data build 25 min, pipeline 8 min), with a bank dashboard for staff and a separate customer app. All customers are synthetic; nothing leaves the machine.

Stage	What runs	Result on 30,000 customers
Data + FR1	7 source systems (71 files) → control checks, crosswalks, de-duplication, quarantine → warehouse → Customer 360 (69 fields, 9 sections) + 63 features	Audit: 8/8 tables and 28,266,495/28,266,495 transactions identical to the system of record; 36/36 realism checks
FR2 categoriser	Boundary rules → merchant lookup → TF-IDF model	Test: 99.0% (seen merchants), 90.4% (merchants never seen in training)
FR3 propensity	Eligibility rules, then per product LightGBM, out-of-fold isotonic calibration, 5-model bootstrap, confidence, TreeSHAP reason codes	Beats the rules baseline on all 6 products; calibration error about 0.01
FR4 impact	DESNZ 2026 well-to-wheel factors; carbon statement (avoided, PCAF financed) + money statement	131,656 customer×product estimates; 48,344 marked *insufficient data* instead of guessed
FR5–FR7	Consent, memory, routing, top 3, human review; template explanation for every recommendation; local LLM rewrite behind guards	5,478 suggestions for 4,824 customers, each at least 2× as likely as average; 292 to human review
FR6 insights	Opportunity by product and region, decision funnel, footprint, 6 behavioural segments, monthly trends	Expected 925 take-ups, 1,237 t CO_2e avoided a year and £418k a year customer savings

2 What changed since M1

- **Richer public data, more checks.** Homes are drawn from DESNZ NEED 2025 (4M real homes: type, age, floor area, metered gas and electricity); car ages and mileage from 9.25M MOT tests (2024); property values from the UK House Price Index; mortgage, loan, card and savings rates from Bank of England quoted rates; age and economic activity by tenure from Census RM201/RM133. Realism checks grew from 24 to 36 and all pass.
- **Scale.** 4,000 → 30,000 customers and 3.8M → 28.3M transactions. Generation, validation and the categoriser stream in parts, so a 14 GB laptop is enough; the GPU cluster is no longer needed.
- **Leaner Customer 360.** Every field was traced to the rule, model or engine that reads it; unread fields were dropped (82 → 69, features 69 → 63). Labels come from the bank's own product system (a green product opened in the window) and match the simulator's truth on all 180,000 customer×product rows.
- **LLM: smaller, local, constrained.** No fine-tuning or distillation. A 1.5B open model (Qwen2.5-Instruct) runs offline and rewrites only the "why + impact" part; the compliance sentence (data date, caveat, investment disclaimer) is fixed text. Our first run passed 0/360 because the model reformatted dates and added promises.
- **Suggestion thresholds.** A fixed 2% propensity floor let in customers no likelier than average. A product is now shown only if its calibrated propensity is at least 2× the product's take-up rate among eligible customers (test lift 2.2–4.3×), confidence is not low, its impact can be estimated, and it avoids at least 100 kg CO_2e a year (Paperless exempt). Suggestions fell from 27,461 to 5,478. Propensity is bounded to 0.1–95%; customer memory is an event log the dashboard writes to.

3 Experiments and results

FR3, test set (customers never seen in training; take-up observed 1 Apr–30 Sep 2026). The oracle scores with the simulator's true probability, including hidden traits, so it is the ceiling for any model.

Product	Take-ups	Rules AUC	Logistic	**LightGBM**	Oracle	Gap closed	PR-AUC	Lift@10%	ECE
GP01 EV Finance	164	0.628	0.726	**0.728**	0.855	44%	0.129	3.0×	0.010
GP02 Green Home Loan	31	0.573	0.597	**0.634**	0.812	26%	0.043	1.9×	0.014
GP03 Solar Finance	56	0.623	0.747	**0.726**	0.906	36%	0.067	3.4×	0.006
GP04 Paperless	122	0.737	0.814	**0.821**	0.827	93%	0.529	4.1×	0.012
GP05 Green Investments	60	0.677	0.764	**0.744**	0.858	37%	0.237	4.3×	0.013
GP06 Green Loans	106	0.625	0.734	**0.709**	0.792	50%	0.090	3.7×	0.005

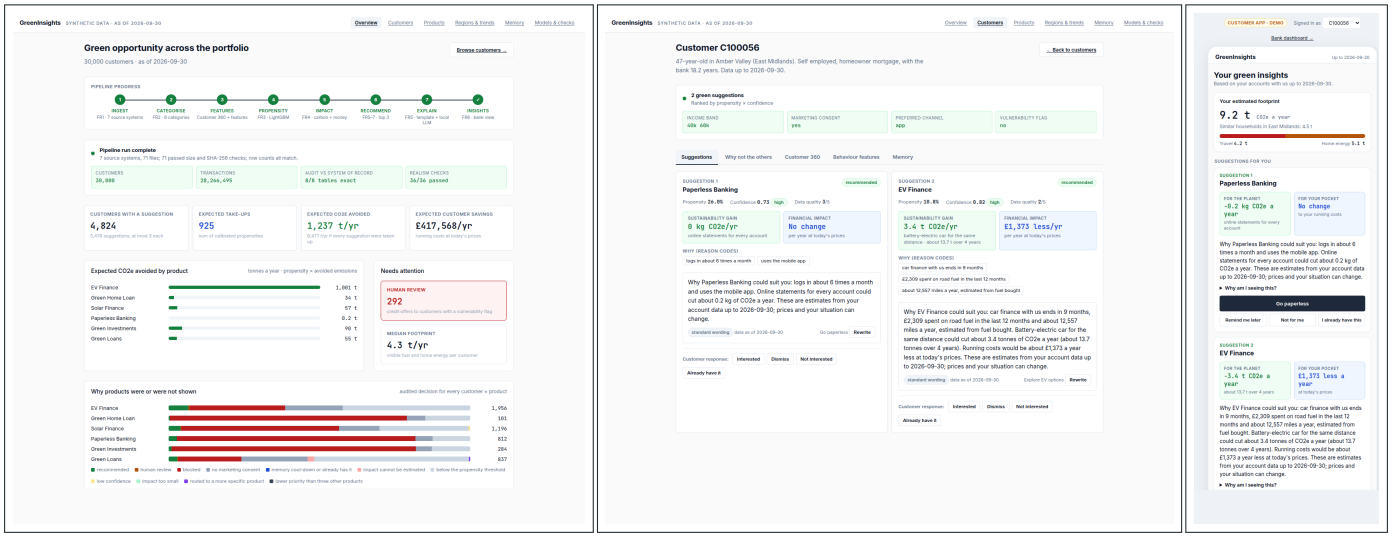
- **Findings.** Calibration matters more than the model family: logistic regression is within 0.04 AUC of LightGBM (better on GP03 and GP05), while the rules baseline is 3–24× worse calibrated (ECE 0.04–0.12 vs 0.005–0.014). Ablations: removing price normalisation cuts GP01 lift from 3.1 to 2.8 and GP02 AUC from 0.66 to 0.63; removing coverage flags cuts GP01 lift to 2.6. Using all 145 inputs instead of the curated 21–40 adds up to 0.023 AUC (GP03) but lets demographic fields drive scores, so we keep the curated, behavioural set.

- **FR2.** Accuracy holds on unseen merchants overall (90.4%), but electricity vs gas from unseen energy suppliers is weak (F1 0.13 and 0.29); on 68k out-of-domain US descriptions, coarse accuracy is 82.1%.
- **FR4 against true usage** (31 Mar snapshot, visible spend only): fuel litres within ±20% for 83% of drivers (r = 0.95); gas and electricity kWh only 39% (r = 0.57 and 0.78), because direct debits smooth bills. Sensitivity: ±20% on km per litre moves GP01 money by 82%; ±20% on solar yield moves GP03 carbon by 40%.
- **FR5.** Templates pass every guard (5,478/5,478). After the redesign the local LLM passes 67.8% of the 360-item evaluation set (60 per product); the rest fall back to the template. Stricter guards now also reject a rewrite that drops a number or turns a 12-month figure into a monthly one, after we found "£1,337 on fuel each month" passing the number check. Reading ease is lower for LLM text (60 vs 72 Flesch).

4 Testing progress

Area	Test	Status
Data	36 checks against Census, DESNZ, DfT/NTS, MOT, BoE, HPI; FR1 audit by table fingerprints and per-transaction hashes; tamper tests (one-penny change, edited narrative, wrong checksum)	36/36; exact; all detected
Automated tests	13 pytest tests: ingestion (8), FR4 golden case worked by hand from DESNZ factors and 30 Sep 2026 prices, insufficient-data path, memory cool-downs (5)	13/13 pass
FR7 rules	On all 5,478 suggestions: none for a blocked product or below its threshold; no solar to renters; nothing without marketing consent; credit offers to vulnerable customers always routed to human review; at most 3 per customer	0 violations
Memory	300 customers, 3 monthly rounds, responses drawn from the true take-up probability	0 products re-shown inside a cool-down; 122 then 197 held back
Fairness	Test-set take-up, mean propensity and selection rate by age, income and region	Low-income bands are rarely selected for EV Finance (0% under £25k), mirroring true take-up

5 Bank dashboard and customer app



Bank dashboard (staff, FR6): portfolio view with pipeline status, expected take-ups and CO_2e by product (left); staff view of one customer with propensity, confidence, data quality, reason codes and the threshold behind every decision (middle). **Customer app** (FR5, right): what the customer sees in their banking app – their footprint against similar households, suggestions in plain language with a "for the planet" and a "for your pocket" statement, "why am I seeing this?", and buttons that write to customer memory. It never shows scores, codes, flags or suggestions awaiting human review.

6 Plan to the final submission

1. **Close measured gaps:** energy kWh from smoothed direct debits (use bill-change dates and annual reconciliation credits); a supplier-name rule list for unseen energy merchants; EV explanations, where only 22% of LLM rewrites pass.
2. **Real-data proofs:** the categoriser on a real UK current-account statement (6,567 lines) and FR1 on a public multi-table bank dataset (1.06M transactions).
3. **Scenario pack** of hand-picked customers with expected outcomes, and a manual review of 60 LLM explanations.
4. **Final:** 3-page report, 4-minute demo of the dashboard (portfolio view → customer → response → re-ranked suggestions), code with setup and run instructions.